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APPLIED RESEARCH

Enhancing Fabric Defect Detection With Attention Mechanisms and Optimized YOLOv8 Framework

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ABSTRACT Fabric defect detection is a critical task in the textile industry, requiring high precision and recall to ensure effective quality control. This study presents an enhanced YOLOv8-based framework that integrates novel attention mechanisms and advanced architectural modules to improve detection accuracy and robustness. The framework incorporates the SimAM attention mechanism within the SPPF module and adopts an optimized Dilation-wise Residual (DWR) structure in the backbone. Comprehensive ablation studies and comparisons with state-of-the-art methods validate the effectiveness of the proposed approach. The enhanced model achieves a mAP₅₀₋₉₅ of 74.3%, outperforming the baseline by 4.7 percentage points, with marked improvements in detecting challenging defect categories. While the framework demonstrates significant advancements, limitations in dataset diversity and computational efficiency are acknowledged. Future work will focus on resource optimization, dataset augmentation, and extending the framework's applicability to other domains.

INDEX TERMS Fabric defect detection, YOLOv8, small target detection, attention mechanism, C2f-R2DWR, SPPF-AEMSim.

I. INTRODUCTION

The textile industry, a historically significant pillar industry worldwide, has made substantial contributions to economic development. In China, textiles serve not only as a key indicator of light industry development but also as a fundamental component of cultural, economic, and social progress. Textiles are extensively used in clothing, home furnishings, and industrial applications. However, their production process is highly susceptible to variations in raw materials, equipment, and technology, leading to defects such as warp and weft misalignment, holes, and stains. These defects compromise both aesthetic appeal and durability, ultimately diminishing product quality, causing economic losses for enterprises, and negatively impacting consumer experiences. Consequently,

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fabric defect detection has become a critical challenge in the textile industry. However, traditional methods such as manual inspection are inherently constrained by operator fatigue, inconsistencies in skill levels, and inefficiency, making them unsuitable for high-speed manufacturing environments [1], [2], [3].

Recent advancements in automated fabric defect detection systems, driven by image processing and machine learning innovations, have revolutionized this field. Early techniques, including edge detection and threshold-based segmentation, demonstrated effectiveness in identifying simple defects but struggled with more complex or small-scale targets due to limited generalization across diverse patterns [4], [5]. Machine learning approaches, such as Support Vector Machines (SVMs), improved detection accuracy by leveraging texture-based handcrafted features. Nonetheless, these methods encountered difficulties in maintaining robustness

within highly variable environments, underscoring the need for adaptive solutions [6]. The emergence of Convolutional Neural Networks (CNNs) marked a significant breakthrough, enabling the automatic extraction of hierarchical features that substantially enhanced accuracy and generalizability [7], [8].

Despite these advancements, single-stage object detection frameworks, particularly those in the YOLO (You Only Look Once) family, face persistent challenges in detecting small-scale defects within intricate and heterogeneous fabric patterns. Although successive iterations, from YOLOv3 to YOLOv8, have introduced lighter architectures and improved small-target detection capabilities in cluttered environments, The experiments presented in this study indicate that there is still room for improvement in detection accuracy [9], [10]. In summary, research on fabric defect detection methods based on deep learning not only has important theoretical value but also holds significant application potential and economic benefits. By continuously optimizing and upgrading the detection model, we aim to enhance the quality of defect detection, which in turn provides robust technical support for the sustainable development of the textile industry.

This study aims to address these challenges by proposing targeted enhancements to the YOLOv8 framework, focusing on improving its capability to detect small-scale defects and complex fabric patterns. The specific objectives are as follows:

- To design a novel C2f-R2DWR (Cross-Scale Fusion and Region-Dependent Weighting Representation) module, integrated into the YOLOv8 feature extraction pipeline, to enhance scale-aware feature representation for small- and medium-sized defects.
- To introduce the SPPF-AMESim (Spatial Pyramid Pooling with Attention Mechanism for Efficient Small-target Detection) module, replacing the original SPPF in YOLOv8, to improve the network's ability to capture discriminative features, particularly for small-target defect detection.
- To validate the proposed method on the Tianchi fabric defect dataset, demonstrating superior performance compared to baseline models and state-of-the-art techniques.

The contributions of this work are twofold. First, the proposed method provides significant potential for improving quality control processes in the textile industry. By enhancing the detection of small-scale defects, it facilitates automated production lines, improves product quality, and reduces manual inspection costs. Second, from an academic perspective, this work offers novel insights into enhancing object detection frameworks, particularly for small-target detection in complex and cluttered environments. The integration of advanced modules, such as C2f-R2DWR and SPPF-AMESim, presents an innovative approach to improving feature extraction and detection accuracy in real-time applications.

Through these contributions, this study advances automated defect detection in the textile industry while offering broader implications for computer vision, particularly in

enhancing the performance of detection frameworks for small targets in complex and variable backgrounds.

II. RELATED WORK

The evolution of fabric defect detection has transitioned from manual inspection to automated methods enabled by advancements in image processing and machine learning. Early techniques, such as edge detection, threshold-based segmentation, and morphological operations, were widely adopted due to their simplicity. However, their reliance on handcrafted features limited adaptability to complex or small-scale defects frequently encountered in textile production [11], [12], [13].

Machine learning methods introduced greater flexibility and improved accuracy. For example, Support Vector Machines (SVMs) combined with texture features enhanced defect detection but required labor-intensive feature engineering and struggled with variations in defect types [14]. The advent of Convolutional Neural Networks (CNNs) marked a paradigm shift, enabling automatic extraction of rich, hierarchical features. Approaches leveraging ResNet and MobileNet backbones achieved robust performance, with higher accuracy in diverse environments [15], [16], [17]. Lightweight models, such as MobileNet, proved advantageous in constrained computational settings, emphasizing the trade-off between performance and efficiency.

Single-stage object detection frameworks, especially the YOLO (You Only Look Once) series, have further advanced defect detection by providing real-time performance and scalability. YOLOv3 achieved a balance between accuracy and efficiency through multi-scale feature learning, while YOLOv5 and YOLOv7 incorporated architectural optimizations, such as enhanced feature pyramid networks and attention mechanisms, to better handle small targets and complex patterns [18], [19], [20]. YOLOv8, the latest in the series, integrates streamlined backbones and lightweight designs, coupled with feature refinement strategies to enhance small-target detection [21].

The incorporation of attention mechanisms has significantly improved the focus on defect-prone regions in fabric images. Modules such as Squeeze-and-Excitation (SE) and Convolutional Block Attention Module (CBAM) have been extensively used to recalibrate features and improve detection accuracy [22], [23]. More recently, transformer-based attention mechanisms and lightweight modules like SimAM (Simple Attention Module) have demonstrated their ability to capture discriminative features with minimal computational overhead [24], [25], [26]. These approaches enhance small-target detection and mitigate challenges posed by cluttered or complex backgrounds.

Despite these advancements, challenges remain in detecting small and obscured defects within intricate fabric patterns. Current models often struggle to balance computational efficiency with detection accuracy in industrial settings. This study builds upon these advancements by

proposing an enhanced YOLOv8 architecture, incorporating novel attention mechanisms and feature extraction modules to address these challenges and further advance the field of fabric defect detection.

III. RESEARCH METHODOLOGY

The research methodology is designed to address the unique challenges of fabric defect detection, including small-scale defects obscured by intricate backgrounds and medium-to-large defects requiring precise boundary identification. To tackle these challenges, the methodology focuses on adapting and enhancing the YOLOv8 framework [21], [23], [25]. The key objectives are:

- Improving multi-scale feature representation,
- Enhancing small-scale defect detection accuracy,
- Maintaining computational efficiency for real-time applications.

To achieve these goals, this study introduces advanced architectural modifications and novel modules. Specifically, the proposed YOLOv8-REE (Refined Enhanced Extensions) model integrates enhanced feature extraction techniques and attention mechanisms to overcome the limitations of conventional object detection methods in fabric inspection tasks. The following subsections provide an in-depth explanation of the proposed innovations, as summarized in Figure 1.

A. BACKBONE: C2f-R2DWR FOR MULTI-SCALE FEATURE EXTRACTION

The detection of medium-to-large fabric defects is enhanced through the proposed **C2f-R2DWR** module, an improved version of the standard C2f backbone in YOLOv8. This module incorporates:

- **Dilation-wise Residual (DWR) Connections:** By leveraging dilated convolutions, DWR connections expand the receptive field with minimal computational cost, enabling the capture of large-scale contextual information.
- **Res2Net Blocks:** These blocks enhance multi-scale feature representation by processing feature channels in finer granularity across multiple scales.

This synergy addresses the limitations of the original C2f backbone, providing a robust balance between computational efficiency and defect detection accuracy.

1) DILATION-WISE RESIDUAL (DWR)

The DWR module addresses the constraints of YOLOv8's original C2f structure, which offers limited multi-scale contextual understanding. The DWR module, inspired by advanced segmentation networks [26], aggregates multi-scale features using residual mechanisms. Its operation is divided into two steps:

- **Region Residualization (RR):** Input features undergo 3×3 convolution, Batch Normalization (BN), and ReLU activation to generate residual feature maps.
- **Semantic Residualization (SR):** Depthwise convolutions with varying dilation rates refine the residual

feature maps, enhancing contextual understanding without redundancy. The refined features are merged via 1×1 convolution and added back to the input.

This design effectively captures diverse contextual information, improving detection performance across complex fabric patterns. Figure 2 illustrates the structure of the DWR module.

2) Res2Net

The **Res2Net** module, as proposed by Gao et al. [27], is integrated to improve multi-scale representation further. Res2Net splits input features into multiple branches, each processed at distinct scales. The outputs are iteratively merged using residual connections, providing a broad receptive field while maintaining computational efficiency.

This architecture captures fine-grained details and global context simultaneously, ensuring robust defect detection across a wide range of sizes and shapes. Figure III-A2 illustrates the Res2Net module.

3) C2f-R2DWR INTEGRATION

The integration of DWR connections and Res2Net blocks into the C2f structure forms the **C2f-R2DWR** module. This architecture leverages the strengths of both components to achieve optimal multi-scale feature representation. After processing features through the DWR module, outputs are refined via Res2Net blocks and concatenated with additional feature maps. This approach enhances detection accuracy for defects of varying sizes and complexities.

The overall architecture of the C2f-R2DWR module is depicted in Figure 4. The red dashed boundary highlights the enhancements introduced in this study.

B. NECK: ENHANCED FUSION WITH SPPF-AEMSIm

Fabric defect detection presents challenges such as subtle size variations and complex background textures. To enhance the detection of small defects, the neck section of YOLOv8-REE integrates the **SPPF-AEMSIm**, an advanced feature aggregation module designed to replace the original Spatial Pyramid Pooling Fast (SPPF) component.

1) KEY COMPONENTS OF SPPF-AEMSIm

- **AEMSIm (Attention-Embedded Module with Similarity Focus):**
 - **SimAM Integration:** The **Similarity Attention Mechanism (SimAM)** dynamically adjusts attention weights based on neuron activity. Neurons corresponding to defect regions receive higher weights, allowing the model to focus on critical areas while suppressing irrelevant background noise.
 - **Enhanced Localization:** This mechanism improves the network's ability to localize subtle features, addressing the characteristics of small fabric defects effectively.

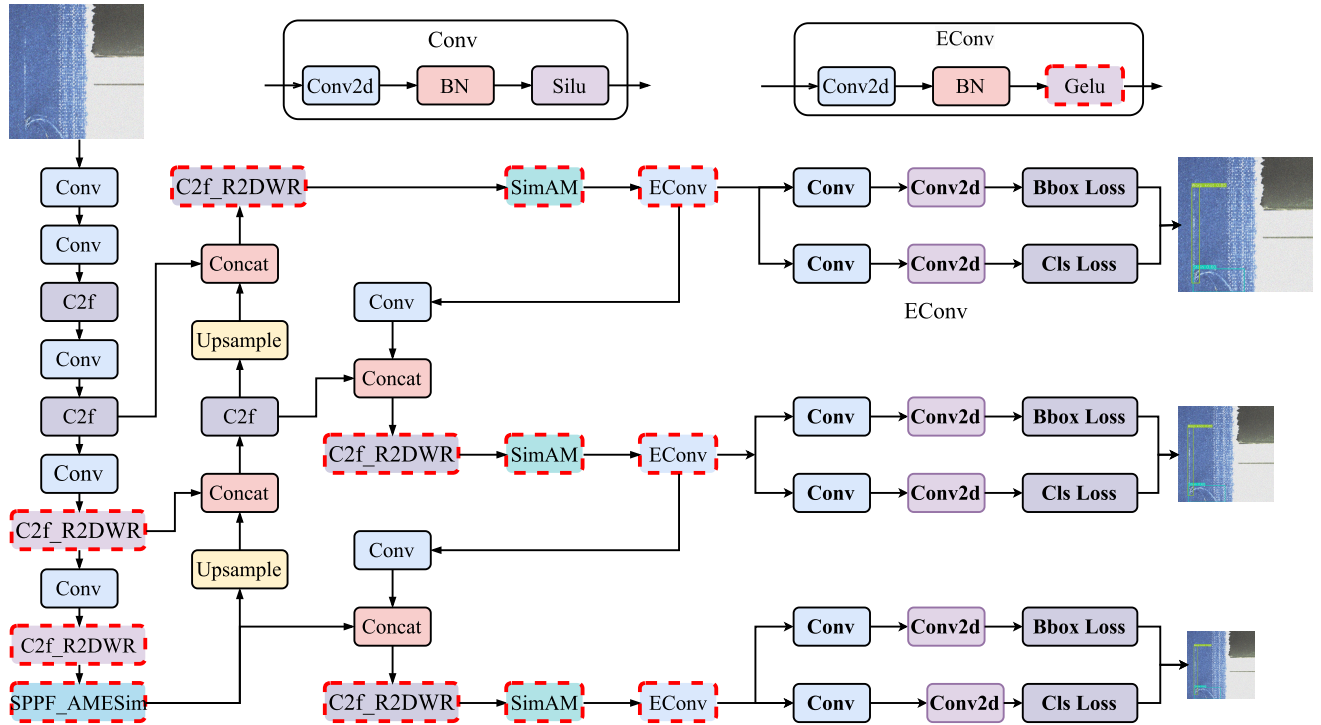


FIGURE 1. Architecture of YOLOv8 for fabric defect detection. The diagram illustrates the YOLOv8 model specifically adapted for textile defect identification, highlighting its key components and enhanced modules in the proposed approach.

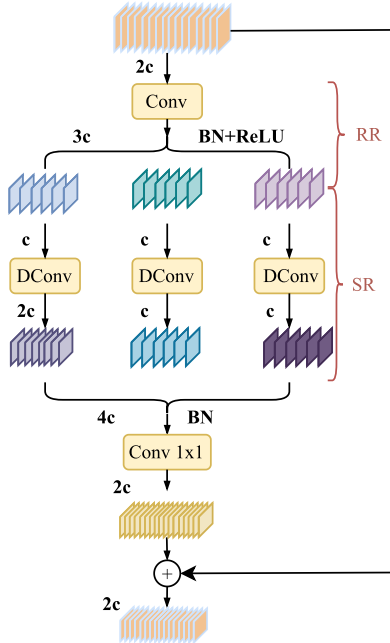


FIGURE 2. DWR module. The DWR module aggregates multi-scale features through residual connections, enhancing defect detection accuracy.

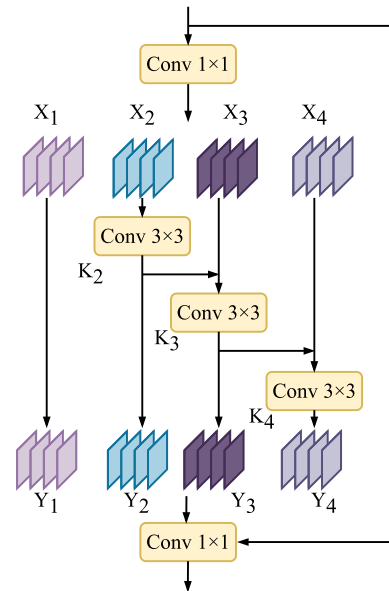


FIGURE 3. Res2Net module. The Res2Net module enhances multi-scale feature representation by splitting feature channels into subspaces, facilitating robust defect detection.

• Custom Enhanced Convolution (Econv):

- Traditional convolutional layers are replaced with **Econv**, a specialized layer that preserves feature details during extreme-value operations.

- By combining outputs from both average and maximum pooling, Econv addresses the SPPF's tendency to overemphasize edge information, ensuring a better balance between edge details and contextual features.

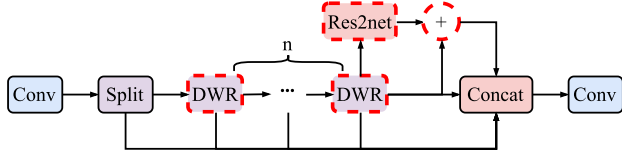


FIGURE 4. C2f-R2DWR module. This module integrates DWR connections and Res2Net blocks within the C2f framework, enabling enhanced multi-scale feature extraction.

• Improved Small Defect Recognition:

- The dual-pooling strategy integrates average and maximum pooling features, resulting in richer and more comprehensive feature representations.
- This approach enhances the detection of small defects, even in scenarios with highly variable backgrounds, significantly improving overall recognition accuracy.

2) SimAM

The SimAM mechanism enhances the model's ability to identify critical features in complex visual data, which is particularly vital for detecting small-scale fabric defects. By effectively guiding attention, SimAM improves the detection accuracy of small, subtle defect patterns [28]. Figure 5 illustrates its framework.

Unlike many attention modules that generate one-dimensional or two-dimensional weights and extend them to channel or spatial dimensions, SimAM introduces a more flexible approach. It estimates three-dimensional weights directly, considering spatial and channel information simultaneously. This innovation avoids the oversimplification of treating regions uniformly, which is common in traditional modules, and enhances feature extraction. It is particularly suitable for significantly enhancing repetitive structures in the texture of fabrics.

SimAM is inspired by the neuroscience concept of spatial suppression, where active neurons suppress neighboring ones to prioritize critical information transmission. It dynamically calculates similarity metrics between targets, assigning higher weights to features that exhibit significant spatial suppression effects. This approach enables precise localization of critical defect features.

The core principle of SimAM is recognizing that not all features contribute equally to defect detection. Like neurons in the human brain that modulate activity based on information richness, SimAM focuses on salient features, ensuring the model attends to the most relevant data [29]. The energy function governing SimAM is as follows:

$$e_t(w_t, b_t, y, x_i) = (y_t - \hat{t})^2 + \frac{1}{M-1} \sum_{i=1}^{M-1} (y_0 - \hat{x}_i)^2 \quad (1)$$

Here, t represents the target neuron, x_i represents other neurons in the feature map X , and $M = H \times W$. After

simplifying and introducing a regularization term λ , the weights w_t and bias b_t are derived as:

$$w_t = -\frac{2(t - \mu_t)}{(t - \mu_t)^2 + 2\sigma_t^2 + 2\lambda}, \quad b_t = -\frac{1}{2}(t + \mu_t)w_t \quad (2)$$

The minimal energy is expressed as:

$$e_t^* = \frac{4(\hat{\sigma}^2 + \lambda)}{(t - \hat{\mu})^2 + 2\hat{\sigma}^2 + 2\lambda} \quad (3)$$

The final output is computed using:

$$\tilde{x} = \text{Sigmoid}\left(\frac{1}{E}\right)X \quad (4)$$

SimAM introduces minimal additional parameters while improving detection accuracy and maintaining high inference speed.

3) SPPF-AEMSim

While SPPF effectively highlights edge information, it often neglects broader contextual details. The enhanced **SPPF-AEMSim** addresses this limitation by incorporating global average and max pooling layers, introducing comprehensive background information to complement edge details. This adjustment enables the network to make more informed decisions, particularly for small-scale fabric defects.

As illustrated in Figure 6, the red dashed components highlight the improvements introduced. By combining global pooling layers and adaptive multi-scale feature fusion, SPPF-AEMSim provides robust spatial attention allocation, significantly improving the detection of subtle and irregular defect features against complex backgrounds. Together with the channel attention weighting mechanism, this enables the network to focus on the edge high-frequency signals of small targets and be able to make more informed decisions.

C. HEAD: DECOUPLED PREDICTION AND ANCHOR-FREE DETECTION

The YOLOv8-REE head employs a **Decoupled Head structure**, which separates regression and classification branches to improve learning efficiency and reduce computational complexity. Unlike traditional anchor-based methods, YOLOv8-REE adopts an **anchor-free detection paradigm**, eliminating the reliance on predefined anchor boxes. Instead, the model directly predicts the center point coordinates and aspect ratios of defects. This paradigm shift reduces computational overhead, simplifies the training process, and improves both detection speed and accuracy.

D. OVERALL ARCHITECTURE AND BENEFITS

The YOLOv8-REE architecture integrates the modifications discussed above into a unified framework, as illustrated in Figure 1. The primary benefits of this model include:

- **Enhanced multi-scale feature extraction:** By incorporating the **C2f-R2DWR** module, the model significantly improves the detection of medium-to-large fabric defects.

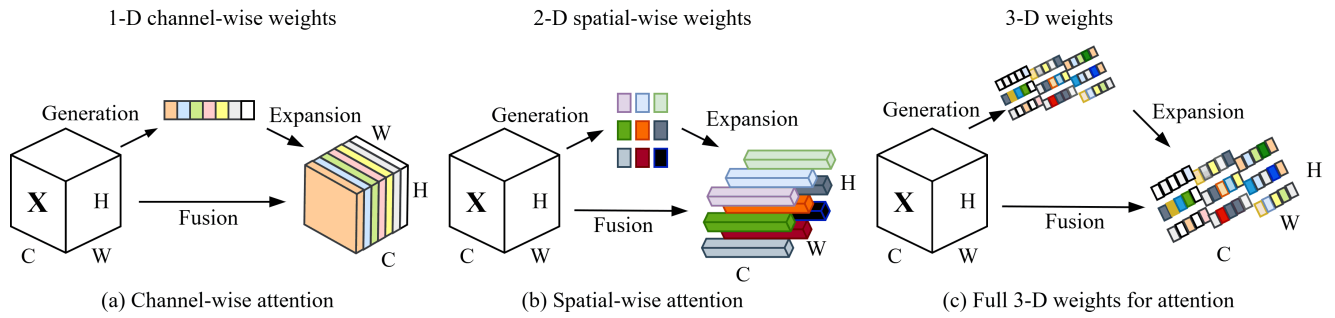


FIGURE 5. SimAM framework.

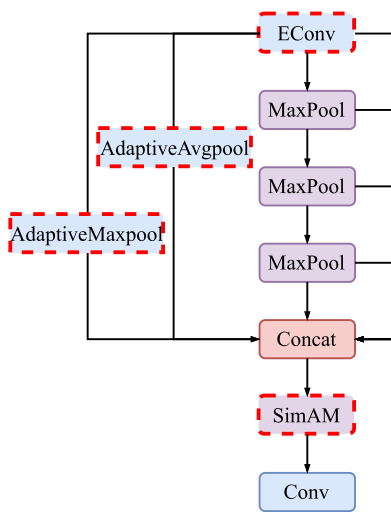


FIGURE 6. SPPF-AEMSim structure.

- **Refined attention mechanisms:** The SPPF-AEMSim module enhances the recognition of small defects within complex backgrounds, addressing challenges in intricate scenarios with high variability.
- **Reduced computational complexity:** Through the use of decoupled head prediction and anchor-free detection, the architecture achieves improved learning efficiency while lowering computational costs.

These advancements enable YOLOv8-REE to achieve exceptional levels of accuracy and efficiency in fabric defect detection, outperforming existing methodologies in terms of precision and robustness.

IV. DATASET

This study utilizes the publicly available fabric defect dataset from the 2019 Guangdong Industrial Intelligence Innovation Competition. The dataset includes 11 defect categories: Thick fiber, Three filaments, Knot, Lint ball, Weft shrinkage, Broken end, Warp knot, Hole, Broken spandex, Stain, and Star jump [30]. These categories represent diverse shapes,

TABLE 1. Distribution of fabric defect categories.

Category	Training	Validation	Test	Total
Thick fiber	886	128	110	1124
Three filaments	1157	165	155	1477
Knot	799	106	87	992
Lint ball	947	104	108	1159
Weft shrinkage	781	92	100	973
Broken end	849	118	123	1090
Warp knot	849	107	106	1062
Hole	1058	156	166	1380
Broken spandex	965	112	90	1167
Stain	1212	156	154	1522
Star jump	829	116	100	1045
Total	8110	1038	1000	10148

sizes, and complexities, mirroring real-world challenges in fabric defect detection.

A. DATASET PROCESSING AND STATISTICS

To ensure consistency and enhance data quality, the original dataset underwent preprocessing, which included:

- **Image Cropping:** All images were resized to a standard dimension of 640×640 pixels, facilitating uniformity during model training and evaluation.
- **Label Cleaning:** Invalid or damaged labels were manually reviewed and removed, resulting in a refined dataset containing 10,148 valid images.

The dataset was partitioned into training, validation, and test sets in an 8:1:1 ratio, resulting in 8,110 training images, 1,038 validation images, and 1,000 test images. Table 1 presents the detailed distribution of defect categories across these subsets, showcasing a relatively balanced class distribution.

B. DATASET BIAS AND LIMITATIONS

Despite its high quality, the dataset exhibits certain biases and limitations:

- **Small Targets:** Defects like lint ball, weft shrinkage, and broken ends are small and often scattered across complex backgrounds, making detection challenging.
- **Large Targets:** Larger defects, such as stains and continuous holes, may overshadow smaller categories during detection, reducing sensitivity to minor defects.

TABLE 2. Training environment and hardware parameter table.

Parameters	Configuration
CPU	Intel Xeon Platinum 8269CY
GPU	NVIDIA A10
Python	3.9
GPU memory size	24G
Operating systems	Ubuntu20.04
Deep learning architecture	PyTorch 1.12.1 + Cuda 11.3

TABLE 3. Experimental parameter settings.

Parameter	Value
Epochs	1000
Initial Learning Rate	0.01
Momentum	0.937
Weight Decay	0.0005
Batch Size	8
Optimizer	Adam
Data Augmentation Strategy	Mosaic
Mosaic Closure Rate	10
IoU Threshold	0.7

- **Class Imbalance:** While the dataset is relatively balanced, some categories contain fewer samples, potentially affecting the detection accuracy for these classes.

To mitigate these issues, the proposed model integrates multi-scale detection capabilities and applies category-specific loss weights during training. However, further data collection and refinement could help address these biases and enhance model robustness.

C. DATASET CONTRIBUTION

By leveraging this comprehensive dataset with rigorous preprocessing and augmentation, this study ensures reliable training and evaluation of the proposed defect detection model. The dataset's diversity and inherent challenges provide a robust benchmark for testing and advancing state-of-the-art detection methods in industrial fabric defect scenarios.

V. EXPERIMENT RESULTS AND DISCUSSION

In this section, we assess the effectiveness of our approach on the dataset.

A. EXPERIMENTAL PARAMETERS

The YOLOv8n model was selected as the baseline due to its lightweight architecture and high detection efficiency, making it particularly suitable for industrial defect detection tasks with complex backgrounds and diverse defect types. Table 2 shows the hardware configuration and environmental settings used during the experimental training phase. Table 3 provides an overview of the key parameters used during the training process.

The training process was designed to balance computational efficiency and model accuracy. The Adam optimizer was employed, leveraging its adaptive learning capability to ensure stable and efficient convergence. A carefully selected learning rate and momentum were used to fine-tune the optimization process.

To further enhance the model's robustness, Mosaic data augmentation was applied with a closure rate of 10. This technique increased sample diversity by combining multiple images into a single composite input, enabling the model to generalize better to variations in defect patterns and complex backgrounds. These carefully chosen parameters collectively ensured a robust training process, yielding a high-performing defect detection model.

Given that the model is trained for up to 1000 epochs, several measures are implemented to prevent and detect overfitting. 1. **Early stopping:** We monitor validation loss, and if it begins to increase while training loss continues to decrease, training is halted, as this indicates overfitting. 2. **Regularization techniques:** Dropout layers are incorporated into the network to reduce the risk of overfitting. 3. **Monitoring training curves:** The training and validation loss curves are closely observed to detect potential overfitting patterns.

B. EVALUATION METRICS

To comprehensively evaluate the performance of the YOLOv8-based fabric defect detection model, we selected three key metrics: precision, recall, and mean Average Precision (mAP). These metrics provide a detailed assessment of the model's ability to accurately and reliably identify fabric defects under real-world conditions.

Fabric defect detection presents unique challenges, such as small-scale defect detection, overlapping patterns, and variations in defect textures and sizes. Therefore, it is crucial to adopt evaluation metrics that measure both the precision of defect identification and the overall robustness of the model. Below, we define the metrics in the context of fabric defect detection:

- **TP (True Positive):** The number of defective fabric instances correctly detected by the model.
- **TN (True Negative):** The number of non-defective fabric instances correctly identified as non-defective.
- **FP (False Positive):** The number of non-defective fabric instances incorrectly classified as defective.
- **FN (False Negative):** The number of defective fabric instances incorrectly classified as non-defective.

1) PRECISION

Precision, also referred to as the positive predictive value, measures the accuracy of defect identification among all samples predicted as defective. In fabric detection tasks, precision reflects the model's ability to minimize false alarms (FP), which is critical in reducing unnecessary inspections in industrial settings. Precision is calculated as:

$$Precision = \frac{TP}{TP + FP} \times 100\% \quad (5)$$

2) RECALL

Recall, or sensitivity, measures the proportion of actual defects correctly identified by the model. For fabric defect detection, recall is particularly important as undetected

defects (FN) can lead to significant downstream issues in production and quality assurance. Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \times 100\% \quad (6)$$

3) AVERAGE PRECISION (AP)

Average Precision (AP) quantifies the area under the precision-recall curve, capturing the trade-off between precision and recall at various thresholds. For defect detection, AP provides insight into the model's consistency in identifying defects across different severities and scales. Mathematically, AP is represented as:

$$AP = \int_0^1 Precision(Recall) d(Recall) \quad (7)$$

4) MEAN AVERAGE PRECISION (mAP)

Mean Average Precision (mAP) extends the AP metric to multiple defect categories, providing a comprehensive measure of the model's performance across all types of fabric defects. mAP is particularly relevant for this study due to the diverse defect categories (e.g., lint balls, stains, holes) with varying complexities and appearances. The mAP metric is calculated as:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \times 100\% \quad (8)$$

Here, AP_i denotes the average precision for the i -th defect category, and N represents the total number of defect categories in the dataset (in this study, $N = 11$).

5) RELEVANCE OF METRICS FOR FABRIC DETECTION

- **Precision and Recall:** Precision ensures that non-defective fabrics are not incorrectly flagged, reducing unnecessary manual inspections. Recall emphasizes identifying as many defective samples as possible, which is critical for maintaining high production quality.
- **AP and mAP:** These metrics assess the model's ability to detect multiple defect types with varying sizes and complexities. High mAP values indicate consistent performance across all defect categories, which is essential in industrial scenarios with diverse defect patterns.

By leveraging these metrics, we ensure a thorough evaluation of the model's accuracy, robustness, and reliability in detecting fabric defects, thereby addressing the practical demands of industrial quality control.

C. COMPARISON OF DIFFERENT ATTENTION MECHANISMS

In this study, we compared five different attention mechanisms—MPCA (Meta-Path and Collaborative Attention) [31], DAttention (Deformable Attention Transformer), MLCA (Mixed Local Channel Attention), SimAM, and LSKA (Large Separable Kernel Attention) [32], [33], [34], [35]. Each mechanism was integrated into the YOLOv8

TABLE 4. Performance of YOLOv8 with different attention mechanisms.

Algorithm	mAP50 (%)	mAP50-95 (%)
YOLOv8+MPCA	93.5	69.7
YOLOv8+DAttention	94.0	69.8
YOLOv8+MLCA	93.8	70.2
YOLOv8+LSKA	93.9	70.1
YOLOv8 (Baseline)	93.3	69.6
YOLOv8+SimAM	94.2	71.0

framework and evaluated against the baseline YOLOv8 model. The same datasets, hardware configuration, and hyperparameter settings were maintained across all experiments to ensure fair and consistent comparisons. The primary evaluation metric was mean Average Precision (mAP), with results presented in Table 4.

1) OBSERVATIONS AND ANALYSIS

From Table 4, it is evident that integrating different attention mechanisms results in varying levels of improvement in mAP performance. These enhancements highlight the effectiveness of attention mechanisms in improving feature extraction, particularly for challenging tasks like fabric defect detection.

- **MPCA, DAttention, MLCA, and LSKA:** These attention mechanisms consistently improved mAP50 and mAP50-95 scores compared to the baseline YOLOv8 model. For instance, the incorporation of MLCA resulted in mAP50 and mAP50-95 values of 93.8% and 70.2%, representing a notable increase from the baseline values of 93.3% and 69.6%. This demonstrates the ability of these mechanisms to emphasize critical regions and reduce false predictions.
- **SimAM Attention Mechanism:** Among all mechanisms tested, SimAM achieved the best results, with mAP50 and mAP50-95 reaching 94.2% and 71.0%, respectively. This represents an improvement of 0.9 and 1.4 percentage points compared to the baseline. The superior performance of SimAM suggests its strong ability to focus on relevant regions while maintaining computational efficiency.

2) PERFORMANCE TRENDS

Figure 7 provides a visual comparison of the mAP50-95 performance across training epochs for the different attention mechanisms. Around the 650th epoch, the SimAM-enhanced YOLOv8 consistently outperforms all other setups, demonstrating superior robustness and convergence in optimizing the detection task.

3) CONCLUSION

The results conclusively indicate that integrating attention mechanisms enhances YOLOv8's performance on fabric defect detection tasks. The SimAM attention mechanism, with its focus on dynamically weighting important regions, exhibited the best overall performance. Therefore, SimAM was selected for further integration into the SPPF to improve the overall detection framework.

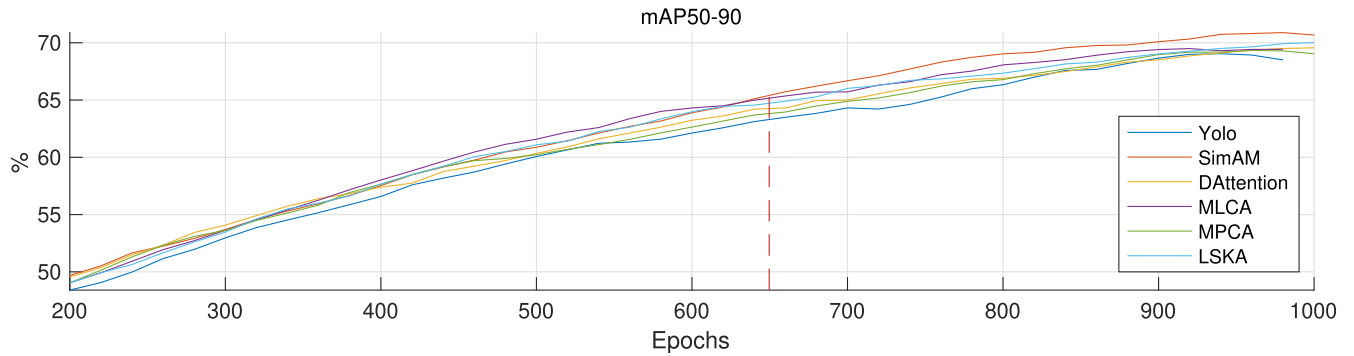


FIGURE 7. mAP50-95 Comparison for attention mechanisms. This figure illustrates the performance trends of YOLOv8 with different attention mechanisms across training epochs. SimAM achieves the highest and most stable results after approximately 650 epochs.

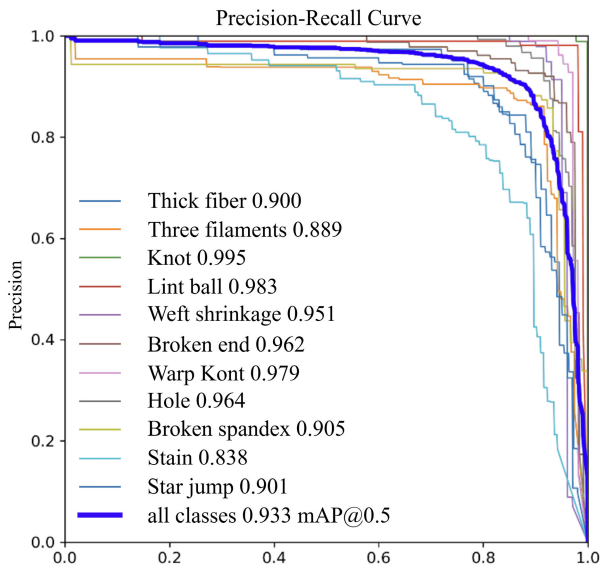


FIGURE 8. Precision-recall curve for YOLOv8 model.

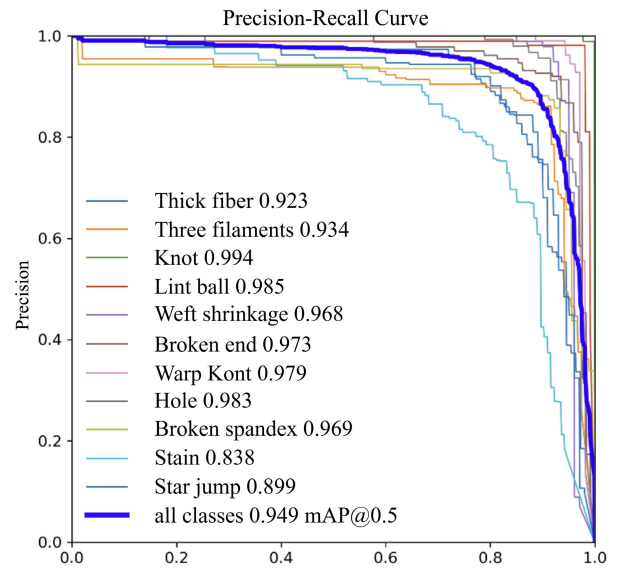


FIGURE 9. Precision-recall curve for our model.

D. ABLATION EXPERIMENTS

Table 5 presents the ablation experiment results, showing the impact of different module configurations. The baseline model (Exp1), based on YOLOv8n, achieved a precision of 91.6%, recall of 89.7%, mAP50 of 93.3%, and mAP50-95 of 69.6%, with a parameter count of 3.01M.

1) IMPACT OF C2f-R2DWR

In Exp2, the integration of DWR within the C2f structure, along with Res2Net, significantly improved the model's performance. Precision increased by 2.5%, mAP50 by 1.3%, and mAP50-95 by 2.2% compared to Exp1. These results highlight the enhanced feature extraction capabilities of this configuration, achieved with a modest increase in parameter count.

2) EFFECT OF SPPF-AEMSim

Exp3 added the SimAM attention mechanism into the SPPF module (forming SPPF-AEMSim). By incorporating global

average pooling, global max pooling, and the GELU activation function, this setup improved recall by 0.8% and boosted mAP50-95 by 3.5% compared to Exp1. This enhancement underscores the importance of capturing fine-grained details for defect detection.

3) COMBINED EFFECT OF AEMSim AND EConv

In Exp4, SimAM and EConv were integrated into the Neck layer, resulting in the best performance. The mAP50-95 reached 74.3%, representing a 4.7% improvement over the baseline. This configuration was particularly effective in detecting complex and small defects, reflecting the synergistic effect of attention mechanisms and convolutional refinements.

Figures 8 and 9 compare the precision-recall curves of the baseline YOLOv8 model and the proposed improved model. The improved model consistently outperformed the baseline, with notable gains in detecting defects such as Thick Fiber

TABLE 5. Experimental results of different configurations.

Exp	C2f-R2DWR	SPPF-AEMSim	AEMSim & EConv	Precision (%)	Recall (%)	mAP50 (%)	mAP50-95 (%)	Params
Exp1	×	×	×	91.6	89.7	93.3	69.6	3,007,598
Exp2	✓	×	×	94.1	89.7	94.6	71.8	3,219,281
Exp3	✓	✓	×	94.4	90.5	94.9	73.1	3,287,910
Exp4	✓	✓	✓	94.4	91.0	94.9	74.3	3,533,714

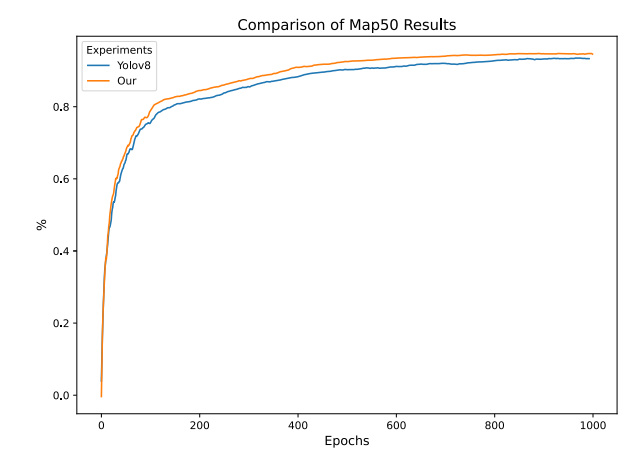


FIGURE 10. Comparison of Map50 results.

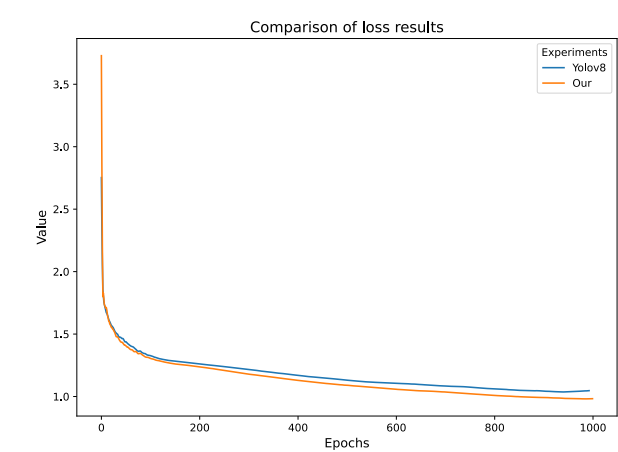


FIGURE 11. Comparison of loss functions.

(+2.3%), Three Filaments (+4.5%), and Broken Spandex (+6.4%).

Figures 10 and Figures 11 respectively compare the mAP50 results and loss function results during the training phase of the baseline model and the improved model proposed in this study. As shown in the figure, the mAP of the proposed model consistently outperforms that of the baseline model and achieves faster convergence.

Figure 12 illustrates the mAP50-95 results across all defect categories. The proposed model consistently achieved higher performance than the baseline, particularly for challenging cases like Cotton Balls, Weft Shrinkage, Broken Ends, and Holes.

4) BALANCING PRECISION AND RECALL

Balancing precision and recall is critical for fabric defect detection due to the diverse characteristics of the dataset. High precision minimizes false positives, reducing unnecessary manual inspections, while high recall ensures that defects are not overlooked, maintaining production standards.

Given the emphasis on detecting rare but critical defects, recall holds slightly higher importance. The proposed model achieved a balanced improvement, with recall rising from 89.7% (Exp1) to 91.0% (Exp4) and precision increasing to 94.4%. This balance underscores the model’s adaptability to the dataset’s requirements and its suitability for real-world deployment in fabric defect detection.

E. QUANTITATIVE COMPARISON WITH RELATED FABRIC DETECTION MODELS

To evaluate the performance and effectiveness of the proposed algorithm, we conducted three sets of comparative experiments. In the first set, the proposed model is compared with the YOLO series algorithms. In the second set, we compare our model with other classical detection models. The third set of experiments compares our model with an enhanced algorithm derived from YOLOv8. Notably, the YOLOv8 FasterNet model replaces the YOLOv8 backbone network with the FasterNet architecture [36]. Table 6, Table 7, and Table 8 summarizes the comparison results, where bold values indicate the best performance.

1) PERFORMANCE IMPROVEMENTS ACROSS METRICS

As shown in Table 6, the proposed model achieves state-of-the-art performance across key evaluation metrics:

① **Precision:** The model attains a precision of 94.4%, outperforming YOLOv8 by 2.8% and matching Literature2. This improvement is attributed to the DWR module, which enhances spatial feature focus on defect-prone regions.

② **Recall:** A recall rate of 91.0% represents a 1.3% gain over YOLOv8, facilitated by the SPPF-AEMSim module, which improves multi-scale feature representation.

③ **mAP50:** The mAP50 score of 94.9% reflects a 1.6% improvement over YOLOv8, achieved through advanced feature fusion layers that maintain high-resolution detail.

④ **mAP50-95:** With a score of 74.3%, the proposed model demonstrates a 4.7% gain over YOLOv8, highlighting the effectiveness of EConv and SimAM modules in refining comprehensive features.

⑤ **Parameter Efficiency:** Despite its competitive performance, the proposed model remains lightweight with

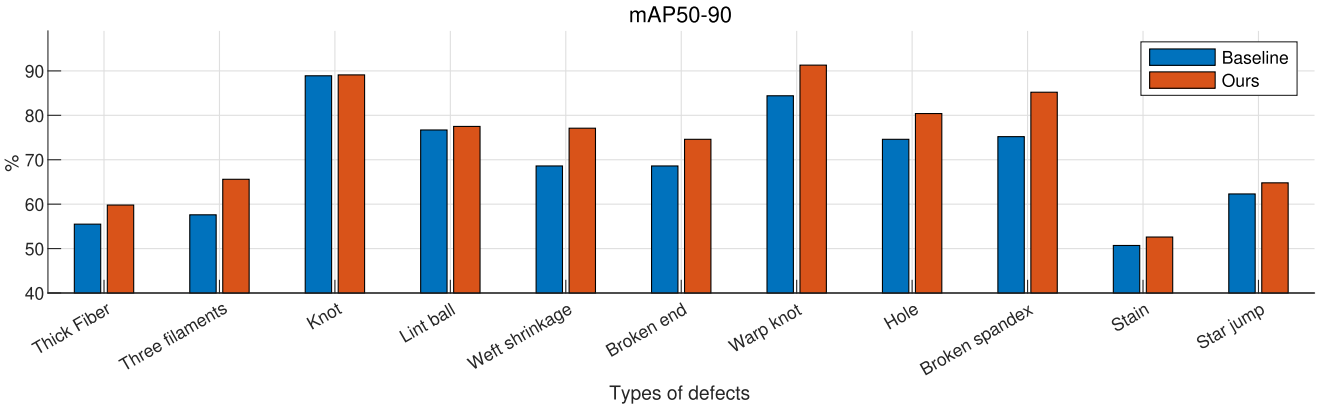


FIGURE 12. Comparison of mAP50-95 results.

TABLE 6. Comparison of detection models.

Model	Precision (%)	Recall (%)	mAP50 (%)	mAP50-95 (%)	Params (M)
YOLOv5	90.5	88.5	92.8	66.2	2.51
YOLOv8n	91.6	89.7	93.3	69.6	3.01
Gold-YOLO [37]	93.4	89.7	94.2	73.0	6.00
YOLOv10 [38]	91.8	90.1	94.3	71.1	2.70
YOLOv11 [39]	93.3	89.8	94.2	69.7	2.60
Ours	94.4	91.0	94.9	74.3	3.53

TABLE 7. Comparison of detection models.

Model	mAP50 (%)	mAP50-95 (%)	Params (M)
Rt-DETR [40]	93.8	70.1	28.5
SSD [41]	78.0	38.9	31.5
Centernet [42]	89.1	50.9	21.9
Ours	94.9	74.3	3.5

TABLE 8. Comparison of detection models.

Model	Precision (%)	Recall (%)	mAP50 (%)	mAP50-95 (%)	Params (M)
YOLOv8-FasterNet	93.1	90.3	93.6	72.3	4.17
AM - YOLOv8 [21]	93.1	90.4	94.7	72.9	3.36
PRC - Light YOLO [20]	94.4	90.5	94.9	73.1	30.50
Ours	94.4	91.0	94.9	74.3	3.53

only 3.53M parameters, making it practical for industrial applications.

2) ANALYSIS OF METRIC CHANGES AND CAUSES

- ① **Precision Improvement:** The integration of the GELU activation function retains extreme feature values, enhancing the model’s ability to distinguish subtle defects.
- ② **Recall Growth:** SPPF-AEMSim ensures accurate detection of small and overlapping defects, reducing false negatives and improving recall rates.
- ③ **Higher mAP Metrics:** The combination of SimAM and DWR strengthens the focus on defect-critical areas, leading to significant enhancements in mAP50 and mAP50-95.
- ④ **Model Efficiency:** By balancing performance with parameter count, the proposed model achieves state-of-the-art results without the computational overhead seen in larger models like Rt-DETR and Literature2.

3) COMPARATIVE STRENGTHS AND WEAKNESSES OF MODELS

- ① **Rt-DETR:** Strong in recall and mAP50-95 but has a high parameter count (28.47M), making it unsuitable for resource-constrained environments.
- ② **YOLOv5 and YOLOv8:** Lightweight and efficient for real-time detection but lack advanced attention mechanisms for precision and recall improvements.
- ③ **Gold-YOLO and Literature2:** Excellent mAP50-95 performance due to sophisticated feature fusion, but high computational demands limit practical deployment.
- ④ **YOLOv10 and YOLOv11:** Compact and competitive, but limited advanced modules restrict their ability to achieve superior accuracy.
- ⑤ **Centernet and SSD:** CenterNet performs well in certain cases by eliminating the computational burden associated with anchor points. However, it is susceptible to

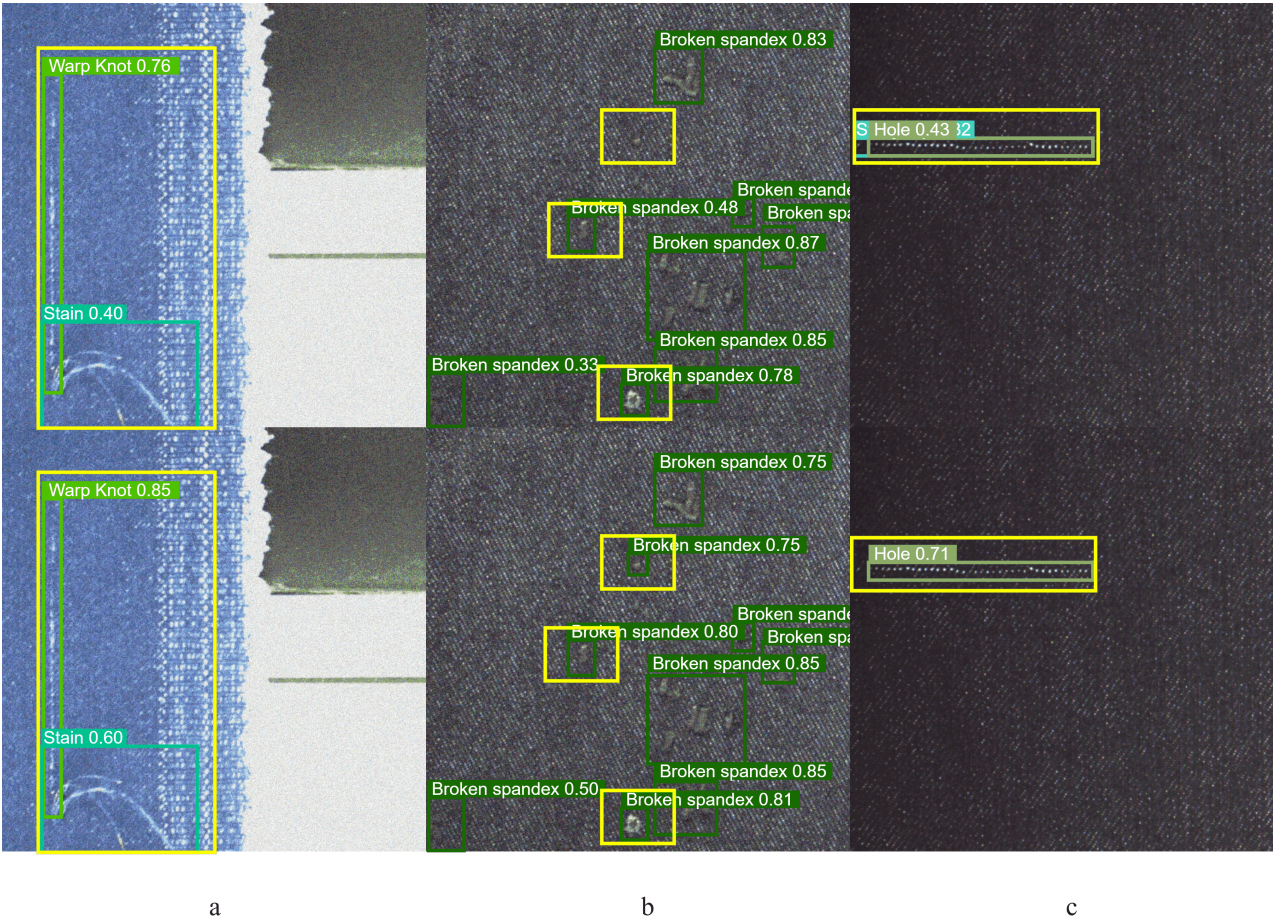


FIGURE 13. Visualization of detection results.

missed detections in dense environments and occluded small objects. SSD requires manual configuration of prior box and aspect_ratio values. Despite utilizing pyramid feature layers, its small object detection accuracy remains inferior to the model proposed in this study.

To verify the robustness and generalizability of the proposed method, we utilized the TILDA dataset for generalization experiments in fabric defect detection. This dataset is specifically designed for surface defect detection in textiles, containing 896 images categorized into four defect types: holes, foreign objects, oil spots, and thread errors. The dataset was divided into a training set (776 images), a validation set (80 images), and a testing set (40 images). We conducted experiments using YOLOv8 and the improved model proposed in this study, comparing their performance(Except for the epoch setting of 200, the hardware, software, and other parameter settings are consistent with this article). The results are presented in Table 9.

As shown in Table 9, the improved model demonstrates superior overall performance compared to YOLOv8.

Based on the results of the three comparative experiments and generalization experiments, our model achieves superior overall detection performance relative to other

TABLE 9. Units for magnetic properties.

data	Model	precision	recall	mAP50	mAP50-95
TILDA	Yolov8	64.8	63.0	73.7	48.3
	Ours	78.0	71.1	77.5	50.7

models. Notably, it offers significant advantages in practical applications, particularly in terms of accuracy and mAP50 improvements.

4) CONCLUSION AND INSIGHTS

The proposed model integrates advanced modules such as DWR and SimAM, achieving state-of-the-art detection performance while maintaining a lightweight design. Future work could focus on further optimization for real-time applications and parameter reduction to enhance industrial scalability.

F. QUALITATIVE ANALYSIS

To highlight the differences in object detection performance between the baseline algorithm and the proposed method, three representative images were selected for visual comparison. The upper portion of each image illustrates the detection

results obtained using the YOLOv8 algorithm, while the lower portion displays the outputs of the enhanced algorithm, as shown in Figure 13.

In Figure 13a, medium-to-large defects are presented. The improved algorithm exhibits a notable enhancement in detecting warp knot defects, increasing the confidence score from 76% (YOLOv8) to 85%. Furthermore, the blue box highlights a stain defect where the confidence score improves from 40% to 60%.

Figure 13b focuses on small-scale defects. Notably, the yellow box at the top indicates a broken spandex defect missed by YOLOv8 but successfully detected by the proposed method. Other small defects also show substantial improvements in detection confidence.

In Figure 13c, a false positive by YOLOv8 is observed. The defect is incorrectly identified as both a stain and a hole, with significantly lower confidence scores compared to the enhanced algorithm. In contrast, the proposed method accurately identifies and classifies the defect. These qualitative results underscore the effectiveness of the improved model in enhancing detection accuracy, thereby aligning with the study's objectives.

VI. CONCLUSION

This study proposed the integration of advanced attention mechanisms into the YOLOv8 framework to enhance the performance of fabric defect detection systems. Five attention mechanisms—MPCA, DAttention, MLCA, SimAM, and LSKA—were systematically evaluated in a controlled experimental setup, ensuring consistency in datasets, hardware configurations, and hyperparameter settings.

The experimental results emphasize the pivotal role of attention mechanisms in improving the detection of complex and subtle defect patterns. Among the tested mechanisms, SimAM emerged as the most effective, achieving the highest mAP50 (94.2%) and mAP50-95 (71.0%), representing improvements of 0.9% and 1.4%, respectively, over the baseline YOLOv8 model. SimAM's ability to dynamically weight features based on their relevance enhanced the model's focus on defect-critical areas, particularly for small and intricate patterns. Additionally, SimAM demonstrated stability across training epochs, making it a robust and reliable choice for fabric defect detection.

The integration of SimAM into the SPPF module led to the development of the refined YOLOv8-REE framework. This enhanced model effectively addresses common challenges in fabric defect detection, such as variations in defect size and complex background textures. It achieves a balance between detection accuracy and computational efficiency, making it suitable for deployment in automated quality control systems. The lightweight architecture of the proposed model ensures scalability, allowing deployment in edge computing environments while maintaining real-time detection capabilities.

The findings of this study have substantial implications for industrial applications. By reducing reliance on manual inspections and enhancing production quality, the proposed

framework offers a practical solution for real-time defect detection in textile manufacturing. Its scalability and adaptability make it a promising candidate for broader adoption in smart manufacturing systems.

VII. LIMITATIONS AND FUTURE WORK

Despite the significant improvements demonstrated by the proposed framework, several limitations remain, which provide directions for future research.

A. LIMITATIONS

- **Dataset Limitations:** The experiments utilized a specific dataset with predefined defect categories. While effective for evaluation, this may limit the model's generalizability to new defect types or diverse materials. Additionally, the dataset does not fully capture real-world conditions, such as varying lighting or fabric textures, which could impact the model's performance in practical scenarios.
- **Resource Requirements:** Although the proposed model maintains a relatively lightweight architecture, the addition of advanced attention mechanisms increases computational complexity. This could hinder deployment in resource-constrained environments, such as edge devices with limited processing capabilities.
- **Evaluation Scope:** The study primarily focused on mAP metrics for performance evaluation, which do not account for other critical factors like inference latency, energy efficiency, or robustness under varying operational conditions.
- **Bias in Defect Handling:** While SimAM outperformed other mechanisms, its enhancements were more pronounced for small or low-contrast defects, potentially introducing biases in the detection of certain defect types.

B. FUTURE WORK

To address these limitations and expand the applicability of the proposed approach, future research should consider the following directions:

- **Enhanced and Diverse Datasets:** Collecting larger and more diverse datasets with varied defect types, materials, and environmental conditions will improve the generalizability of the model. Synthetic data augmentation and domain adaptation techniques can further complement these efforts.
- **Resource Optimization:** Exploring techniques like model pruning, quantization, or neural architecture search can reduce computational demands, enabling the deployment of the model on edge devices and in low-resource environments.
- **Comprehensive Evaluation Metrics:** Including additional metrics, such as inference time, energy consumption, and robustness to variations in environmental conditions, will provide a more holistic assessment of the model's practicality for industrial use.

- **Hybrid Attention Mechanisms:** Combining different attention mechanisms to exploit their complementary strengths may further improve feature extraction and detection performance. Adaptive frameworks that dynamically select attention mechanisms based on input characteristics could also be explored.
- **Cross-Domain Applications:** Extending the proposed approach to other object detection tasks, such as medical imaging, autonomous driving, or industrial anomaly detection, will validate its generalizability and effectiveness across domains.

By addressing these challenges and pursuing the proposed future directions, the framework can be refined to achieve greater robustness, versatility, and real-world impact. This will facilitate its adoption in diverse industrial applications, further advancing automated defect detection technologies.

REFERENCES

- [1] J. Liu, C. Wang, H. Su, B. Du, and D. Tao, "Multistage GAN for fabric defect detection," *IEEE Trans. Image Process.*, vol. 29, pp. 3388–3400, 2020.
- [2] B. Lu, M. Zhang, and B. Huang, "Deep adversarial data augmentation for fabric defect classification with scarce defect data," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–13, 2022.
- [3] N. Alruwais, E. Alabdulkreem, K. Mahmood, R. Marzouk, M. Assiri, A. A. Abdelmageed, S. Abdelbagi, and S. Drar, "Hybrid mutation moth flame optimization with deep learning-based smart fabric defect detection," *Comput. Electr. Eng.*, vol. 108, May 2023, Art. no. 108706.
- [4] D. Powell, M. C. Magnanini, M. Colledani, and O. Myklebust, "Advancing zero defect manufacturing: A state-of-the-art perspective and future research directions," *Comput. Ind.*, vol. 136, Apr. 2022, Art. no. 103596.
- [5] K. T. Chaka, A. A. Shiferaw, and S. T. Sharew, "Inspection of cotton woven fabrics produced by Ethiopian textile factories through a real-time vision-based system," *J. Natural Fibers*, vol. 20, no. 2, Nov. 2023, Art. no. 2286615.
- [6] Y. Huang, J. Jing, and Z. Wang, "Fabric defect segmentation method based on deep learning," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–15, 2021.
- [7] F. G. Yaşar Çıklaçandır, S. Utku, and H. Özdemir, "Determination of various fabric defects using different machine learning techniques," *J. Textile Inst.*, vol. 115, no. 5, pp. 733–743, May 2024.
- [8] D. Harel, R. Yerushalmi, A. Marron, and A. Elyasaf, "Categorizing methods for integrating machine learning with executable specifications," *Sci. China Inf. Sci.*, vol. 67, no. 1, Jan. 2024, Art. no. 111101.
- [9] N. Kuznetsova, Z. H. Sagirova, I. Dhif, D. Gognieva, N. Gogiberidze, P. Chomakhidze, and P. Kopylov, "Determination of left ventricular diastolic dysfunction using machine learning methods," *Eur. Heart J.*, vol. 42, no. 1, Oct. 2021, Art. no. ehab724.3051.
- [10] A. Zahra, M. Amin, F. E. A. El-Samie, and M. Emam, "Efficient utilization of deep learning for the detection of fabric defects," *Neural Comput. Appl.*, vol. 36, no. 11, pp. 6037–6050, Apr. 2024.
- [11] Y. Kahraman and A. Durmuşoğlu, "Deep learning-based fabric defect detection: A review," *Textile Res. J.*, vol. 93, nos. 5–6, pp. 1485–1503, Mar. 2023.
- [12] W. Liqun, W. Jiansheng, and W. Dingjin, "Research on vehicle parts defect detection based on deep learning," *J. Phys., Conf. Ser.*, vol. 1437, no. 1, Jan. 2020, Art. no. 012004.
- [13] Y. Xie, W. Hu, S. Xie, and L. He, "Surface defect detection algorithm based on feature-enhanced YOLO," *Cognit. Comput.*, vol. 15, no. 2, pp. 565–579, Mar. 2023.
- [14] K. Patel, V. Patel, V. Prajapati, D. Chauhan, A. Haji, and S. Degadwala, "Safety helmet detection using YOLO V8," in *Proc. 3rd Int. Conf. Pervasive Comput. Social Netw. (ICPCSN)*, Jun. 2023, pp. 22–26.
- [15] B. Wei, K. Hao, L. Gao, and X.-S. Tang, "Detecting textile micro-defects: A novel and efficient method based on visual gain mechanism," *Inf. Sci.*, vol. 541, pp. 60–74, Dec. 2020.
- [16] Y. Liu, Y. Liu, P. Zhang, Q. Zhang, L. Wang, R. Yan, W. Li, and Z. Gui, "Spark plug defects detection based on improved faster-RCNN algorithm," *J. X-Ray Sci. Technology: Clin. Appl. Diagnosis Therapeutics*, vol. 30, no. 4, pp. 709–724, Aug. 2022.
- [17] M. Chen, F. Bai, and Z. Gerile, "Special object detection based on mask RCNN," in *Proc. 17th Int. Conf. Comput. Intell. Secur. (CIS)*, Nov. 2021, pp. 128–132.
- [18] J. Jing, D. Zhuo, H. Zhang, Y. Liang, and M. Zheng, "Fabric defect detection using the improved YOLOv3 model," *J. Engineered Fibers Fabrics*, vol. 15, Jan. 2020, Art. no. 1558925020908268.
- [19] G. Lin, K. Liu, X. Xia, and R. Yan, "An efficient and intelligent detection method for fabric defects based on improved YOLOv5," *Sensors*, vol. 23, no. 1, p. 97, Dec. 2022.
- [20] B. Liu, H. Wang, Z. Cao, Y. Wang, L. Tao, J. Yang, and K. Zhang, "PRC-light YOLO: An efficient lightweight model for fabric defect detection," *Appl. Sci.*, vol. 14, no. 2, p. 938, Jan. 2024.
- [21] Z. Xu, Y. Bao, and B. Tian, "Improved YOLOv8 garment sewing defect detection method based on attention mechanism," *J. Meas. Eng.*, vol. 12, no. 4, pp. 706–721, Dec. 2024.
- [22] J. Terven, D.-M. Córdova-Esparza, and J.-A. Romero-González, "A comprehensive review of YOLO architectures in computer vision: From YOLOv1 to YOLOv8 and YOLO-NAS," *Mach. Learn. Knowl. Extraction*, vol. 5, no. 4, pp. 1680–1716, Nov. 2023.
- [23] M. Sohan, T. S. Ram, and C. V. R. Reddy, "A review on YOLOv8 and its advancements," in *Proc. Int. Conf. Data Intell. Cognit. Informat.*, Jan. 2024, pp. 529–545.
- [24] G. Jocher, "YOLOv5 by ultralytics," Zenodo, Aug. 2020, doi: 10.5281/zenodo.3983579. [Online]. Available: <https://github.com/ultralytics/yolov5>
- [25] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new State-of-the-Art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.
- [26] H. Wei, X. Liu, S. Xu, Z. Dai, Y. Dai, and X. Xu, "DWRSeg: Rethinking efficient acquisition of multi-scale contextual information for real-time semantic segmentation," 2022, *arXiv:2212.01173*.
- [27] S.-H. Gao, M.-M. Cheng, K. Zhao, X.-Y. Zhang, M.-H. Yang, and P. Torr, "Res2Net: A new multi-scale backbone architecture," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 2, pp. 652–662, Feb. 2021.
- [28] L. Yang, R. Zhang, L. Li, and X. Xie, "SimAM: A simple, parameter-free attention module for convolutional neural networks," in *Proc. Int. Conf. Mach. Learn.*, Jul. 2021, pp. 11863–11874.
- [29] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [30] Tianchi, "Smart diagnosis of cloth flaw dataset," Guangdong Industrial Wisdom Innovation Competition, Guangdong Provincial People's Government, 2020. [Online]. Available: <https://tianchi.aliyun.com/dataset/dataDetail?dataId=79336>
- [31] H. Lu, K. N. Plataniotis, and A. N. Venetsanopoulos, "MPCA: Multilinear principal component analysis of tensor objects," *IEEE Trans. Neural Netw.*, vol. 19, no. 1, pp. 18–39, Jan. 2008.
- [32] Z. Xia, X. Pan, S. Song, L. E. Li, and G. Huang, "Vision transformer with deformable attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4784–4793.
- [33] D. Wan, R. Lu, S. Shen, T. Xu, X. Lang, and Z. Ren, "Mixed local channel attention for object detection," *Eng. Appl. Artif. Intell.*, vol. 123, Aug. 2023, Art. no. 106442.
- [34] K. W. Lau, L.-M. Po, and Y. A. U. Rehman, "Large separable kernel attention: Rethinking the large kernel attention design in CNN," *Expert Syst. Appl.*, vol. 236, Feb. 2024, Art. no. 121352.
- [35] J. Chen, S.-H. Kao, H. He, W. Zhuo, S. Wen, C.-H. Lee, and S.-H.-G. Chan, "Run, Don't walk: Chasing higher FLOPS for faster neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 12021–12031.
- [36] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
- [37] C. Wang, W. He, Y. Nie, J. Guo, C. Liu, K. Han, and Y. Wang, "Gold-YOLO: Efficient object detector via gather-and-distribute mechanism," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2023, pp. 51094–51112.
- [38] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.
- [39] G. Jocher, J. Qiu, and A. Chaurasia, "Ultralytics YOLO," Improved Version, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>

- [40] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 16965–16974.
- [41] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot MultiBox detector," in *Proc. ECCV*, Amsterdam, The Netherlands, Oct. 2016, pp. 21–37.
- [42] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.



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